

Research Article

Principal component analysis for Imidazo[4,5-*b*]pyridine linear and nonlinear optical properties descriptionO.D. Marbello , L. De Boni, C.R. Mendonça ^{*} *Instituto de Física de São Carlos, Universidade de São Paulo, CP 369, 13560-970, São Carlos, SP, Brazil*

A B S T R A C T

The two-photon absorption efficiency in materials depends on numerous factors, including the type of material, electronic structure, solvent, thermal and chemical properties, etc. In addition, when analyzing optical systems using nonlinear spectroscopy, a significant amount of data and variables that are not easily interpretable or correlated are involved. Quantum chemical calculations and other methods have been used to describe the nonlinear optical properties of materials. However, such approaches require a high level of understanding of the material's electronic structure, molecular geometry, etc. Yet, the experimental conditions can significantly affect the two-photon absorption cross-section. Some studies use multivariate analysis techniques to simplify the treatment and interpretation of large data sets without a high computational cost. Principal Component Analysis (PCA) is a common method for reducing and exploring large databases. In addition, PCA results can be interpreted as a new dependent variable for regression models. This paper uses the PCA as a multivariate analysis technique to simplify and interpret the two-photon absorption and photophysical properties obtained in the nonlinear characterization of six Imidazo[4,5-*b*]Pyridine, as a case of study. Linear regression using principal component scores was used to investigate the dependence of the two-photon absorption cross-section intensity on photophysical parameters (descriptor variables). This was achieved by projecting the values of each variable into the component space and comparing them to the values obtained for the two-photon absorption cross-section. The results revealed that PCA can reproduce the experimental results separating the sample studied based on their electronic structure, with a cumulative variance higher to 80 %. The regression model indicates that the first and second excited state transition dipole moments, permanent dipole moment, and fluorescence quantum yield dominate the behavior of the two-photon absorption cross-section, showing high correspondence with the experimental data.

1. Introduction

Two-photon absorption (2PA) is a nonlinear optical phenomenon in which an electronic system is excited by the simultaneous absorption of two photons, with their combined energies matching the transition energy [1–3]. The probability of a material absorbing two photons is proportional to the square of the excitation source intensity [3,4]. Such a relationship results in highly localized excitation. Consequently, materials exhibiting significant 2PA have been widely used in several fields, including microscopy, microfabrication, telecommunications, and photodynamic therapy [5–7].

Several materials can be used as a nonlinear medium, including organic molecules with extended π -conjugated systems [5], nano-materials such as quantum dots [8], nanorods, graphene [9,10], and inorganic materials like semiconductors and glasses [9,11–13]. The efficiency of 2PA in materials is directly linked to various interrelated factors, including their structural and electronic properties, as well as manufacturing processes [14,15]. Organic molecules with high electronic conjugation have been shown to present a high 2PA cross-section

(2PACS) [16–18]. Also, the presence of electron acceptor or donor groups leads to a further increase in the 2PACS [19–21]. Therefore, with advancements in molecular design techniques, it is possible to obtain numerous structures with different peripheral groups and geometries, each with unique optical properties [22–24], whose 2PA spectrum is obtained by using transmission (Z-scan) or fluorescence-based methods. As a rule of thumb, such experimental data are manually analyzed considering the molecular structure and strength of the electron donor/acceptor peripheral groups, with the aid of phenomenological approaches and quantum chemical calculations, to trace a structure-property relationship.

Although this approach has been used for quite some time, it becomes challenging when there are many compounds to analyze or when the differences between their nonlinear responses are subtle. In this context, multivariate analysis techniques have been employed to simplify the treatment and interpretation of data sets. One commonly used method for reducing and exploring large databases is Principal Component Analysis (PCA) [25–27]. PCA transforms a space of p variables and n observables into a subspace of X_n linear combinations, which

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results from the covariance matrix of the p correlated variables [28,29]. Hence, PCA performs a change of basis to retain the highest variance of the original data, making it easier to handle and interpret large databases [28,29].

The Principal Component Analysis (PCA) method has been widely implemented across multiple fields of study, particularly for the analysis of spectral data when working with large datasets [30]. Additionally, PCA has proven to be an effective tool for variable selection, enabling the identification of key parameters that influence specific properties. In the field of optical materials, PCA has been employed to determine the role of various molecular and structural parameters in defining the optical properties of materials [28,30,31]. In this way, this work explores the use of PCA as an alternative to describe and interpret the experimental 2PA spectrum in organic compounds. Specifically, we employed PCA in the 2PA cross-section spectrum of six imidazo[4,5-*b*]pyridine derivatives with different electron-donating and accepting groups, which have been experimentally characterized previously [32]. In addition, PCA was used to describe the experimental data and to analyze the interdependence between photophysical parameters in the component space and their correlation with 2PA cross-section values. Such analysis and the interpretation of the results were compared to the one obtained in Ref. [32], carried out using the standard human/manual analyses.

2. Materials and methods

2.1. Imidazopyridines optical properties and two-photon absorption

In a previous study [32], we investigated six derivatives of imidazo[4,5-*b*]pyridine [32], whose molecular structures are presented in Fig. 1. Each compound consists of an imidazo[4,5-*b*]pyridine core with a methoxy-ethoxymethyl group in the N-3 position and a phenyl group attached to the C-2 position via a triple bond. The derivatives have distinct structures, characterized by the electron-acceptor or electron-donor groups linked in the *meta* or *para* positions of the imidazole core or the phenyl.

The molecules can be classified into two groups based on the nature of their peripheral groups: Donor (D) or Acceptor (A). These groups are attached to the imidazopyridine core (Imp). The A-Imp-A group consists of derivatives containing only electron acceptors, such as bromine (Br), fluorine (F), and trifluoromethyl (CF₃) groups linked to the Imp core. The A-Imp-D molecules feature a combination of the methoxy group, which is an electron donor, with halogenic acceptor groups like bromine (Br) and chlorine (Cl).

The 2PA cross-section spectrum of such compounds was determined in Ref. [32] using the open-aperture Z-scan technique [33,34], in the spectral interval from 480 to 800 nm. In such a study, the nonlinear optical properties were analyzed by interpreting the results considering the strength of the electron donor/acceptor groups, using the sum-over-essential-state (SOES) approach, and performing quantum chemistry calculations. Specifically, Density Functional Theory (DFT) calculations were performed using the Gaussian 09 package [35], to obtain the frontier molecular orbitals (FMO) [36], ground state (μ_{00})

permanent dipole moments, and the cavity volume of imidazo[4,5-*b*]pyridine derivatives. Additionally, in Ref. [32], linear optical measurements, such as linear absorption, fluorescence, solvatochromism, fluorescence quantum yield, etc, have been obtained, some of which will be used in this work for the PCA.

2.2. PCA model construction

The PCA model was performed using the R software within the Rstudio programming environment. Specifically, FactoMineR and Factoextra packages were employed to obtain the covariance matrix and explain variance distribution. The initial step of the PCA model construction involved standardizing the original data. This is achieved by subtracting each value by its arithmetic mean and dividing each by its standard deviation. This standardization was necessary because the principal components (PCs) are identified based on the directions of greatest variation (maximum variance). In a non-standardized dataset, the calculation of principal components (PCs) could be influenced by the variable with the highest magnitude [31].

The number of principal components (PC) used in the PCA model was chosen based on the *scree test*, which represents the percentage of variance explained as a function of the number of PCs [37]. A secondary criterion for selecting the PCs was ensuring that the cumulative variance accounted for more than 80 % of the data variability. After this, each model was interpreted using the Scores or the Loadings plots [29,38]. The Score plot shows the spatial distribution of individuals in the component space, indicating the distance of each observation from the primary axes. On the other hand, the Loadings plot, also known as $\cos^2(\theta)$ plot, allows to observe the correlation between the descriptor variables in terms of the angular relationship between them. To facilitate the graphics interpretation and correlate the variables in the components space, sample structure (A-Imp-A and A-Imp-D) was included as a categorical variable, while a 90 % confidence ellipse was introduced.

3. Results

Fig. 2 shows the 2PA spectrum for the six imidazopyridine derivatives used in this work, obtained from Ref. [32]. The spectra were stacked in the vertical axes to aid visualization. The 2PA spectral features present a main band around 520 nm with a monotonical decrease as the excitation wavelength increases. Additionally, the A-Imp-D samples display a second 2PA band of around 650 nm. Notably, the magnitude of the 2PA bands also varies with the molecular structure. As mentioned before, usually, such data is manually analyzed/interpreted with the aid of theoretical models and quantum chemistry calculations.

A first, PCA was performed to observe the sample categorization based on the 2PA cross-section spectrum of the imidazopyridine derivatives shown in Fig. 2. As a preprocessing treatment, the 2PA spectra for all the molecules were centralized on the same spectral range. That is possible because the experimental system employed in the nonlinear characterization Ref. [32] uses steps of 10 nm. In addition, a data exclusion was implemented using the same wavelengths for all molecules in the PCA analysis. The first two principal components

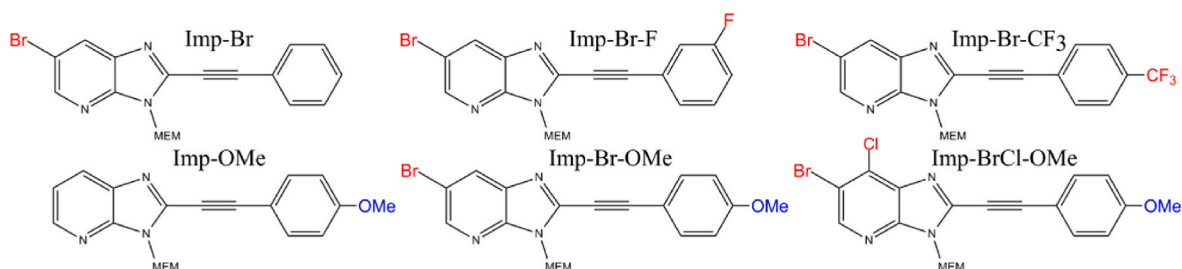


Fig. 1. Chemical structures of imidazo[4,5-*b*]pyridine derivatives.

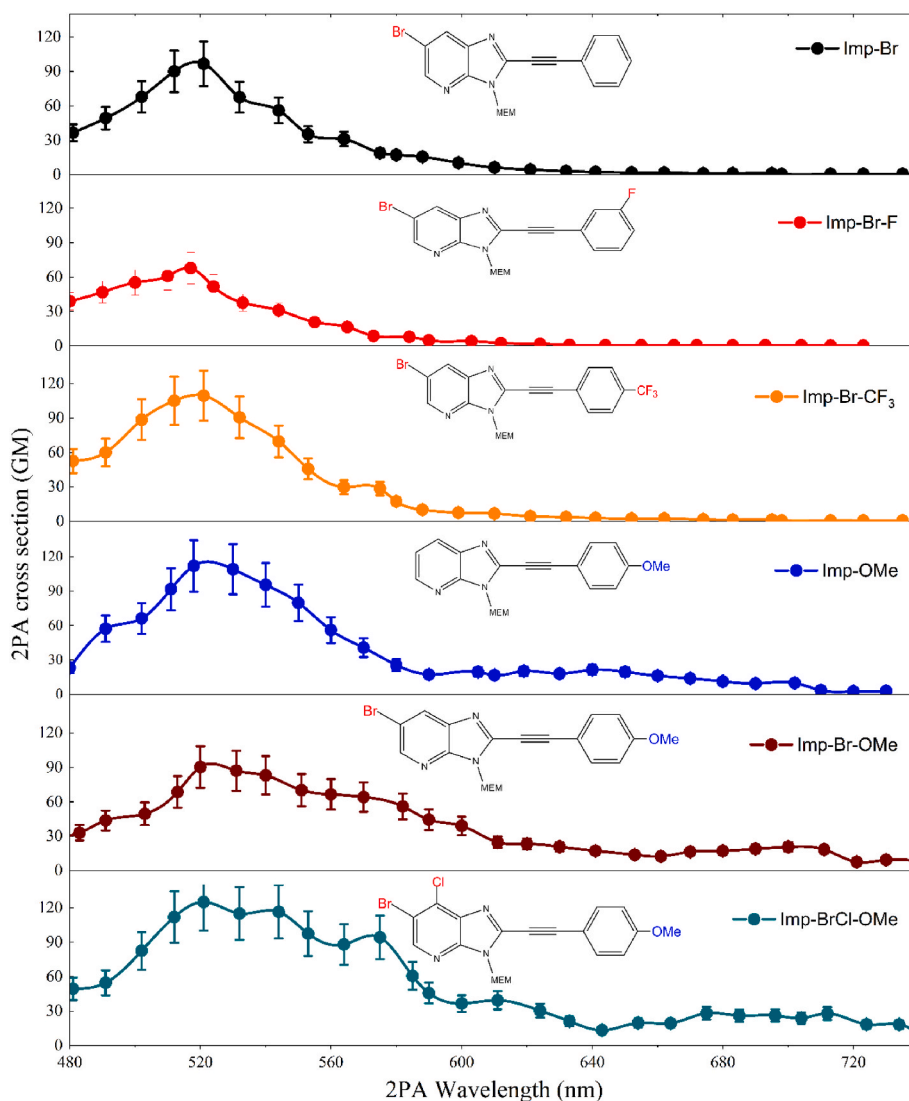


Fig. 2. 2PA cross-section spectra for the six imidazo[4,5-b]pyridine studied in Ref. [32].

collectively explained 93 % of the variance, with the first principal component (PC1) explaining 80.2 % and the second principal component (PC2) explaining 13.6 %. Moreover, to support their selection, the Loading plot, shown in Fig. 3, depicts the 2PA cross-section spectra description in the component space. It is noticeable that up to the first two principal components, the spectral behavior of the original data is partially reproduced. However, a random distribution is observed for

the components PC3 and PC4, with no similarity to the original distribution, indicating noise description. In particular, the spectral region with a high contribution is between 520 and 600 nm for PC1, while for PC2 it ranges from 480 to 570 nm.

The analysis of the principal component scores (see Fig. 4) showed a clear separation and clustering of the samples, introducing a 90 % confidence ellipse and sample structure as a categorical variable.

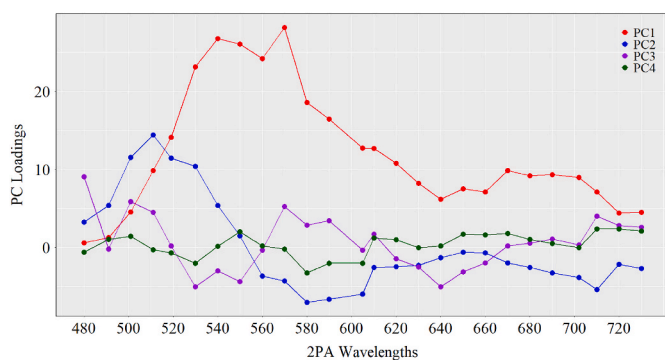


Fig. 3. Loading plot for the six imidazo[4,5-b]pyridine derivatives 2PA cross-section spectra characterized.

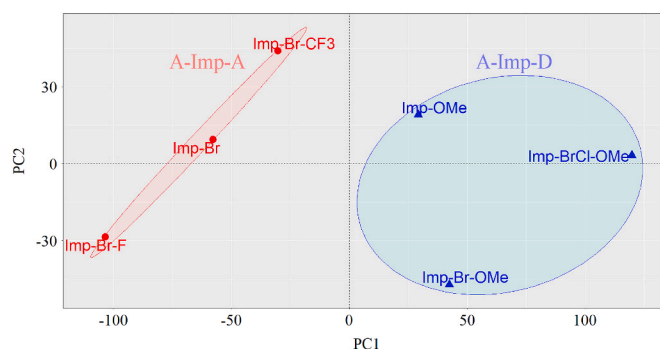


Fig. 4. Score plot for the six imidazo[4,5-b]pyridine derivatives, categorized according to the electronic structure.

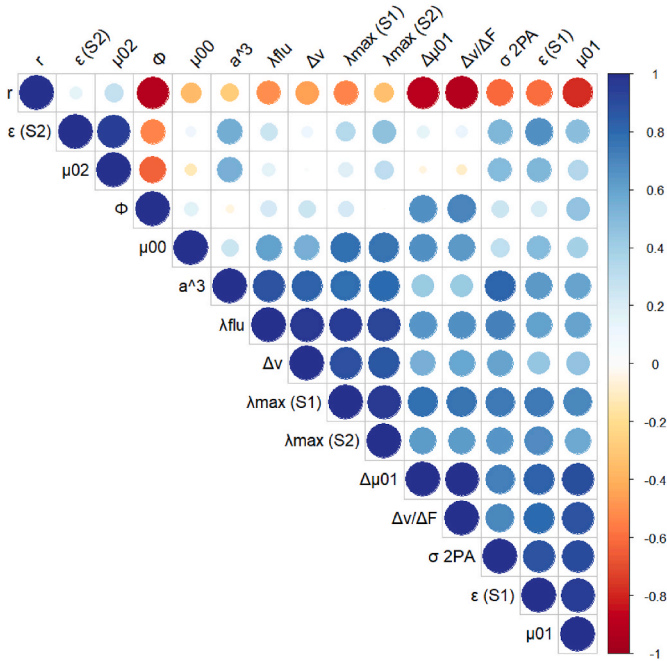


Fig. 5. Correlation matrix of all the molecules' linear and nonlinear properties descriptors.

As seen in Fig. 4, A-Imp-A molecules show a negative correlation with the PC1 and are mainly distributed along the PC2 axes. The Imp-Br-CF₃ and Imp-Br-F samples were highly described by PC2 in positive and negative directions, respectively. This indicates a positive and negative correlation with the 2 PA cross-section in the spectral regions described by this component (480–570 nm). On the other hand, the A-Imp-D samples are equally distributed along the vertical axes with a positive correlation with the PC1. Specifically, the Imp-BrCl-OME sample shows a higher positive correlation with the PC1, indicating better behavior in terms of the 2PA cross-section value. All these results are consistent with the interpretation of the 2PA cross-section spectra (Fig. 2) described in Ref. [32].

Another important consideration is that, for large datasets or those containing variables with multiple correlations, determining the influence of descriptive variables on a target variable becomes highly challenging. Variables with multicollinearity carry redundant information, and relying solely on conventional techniques makes it difficult to distinguish the individual effects of each variable and their specific relationship to a target variable. In our case, Fig. 5 illustrates the correlation matrix among all parameters, including the 2PA cross-sections.

Note that multiple positive and negative correlations can be observed between the 2PA cross-sections (σ_{2PA}) and various photo-

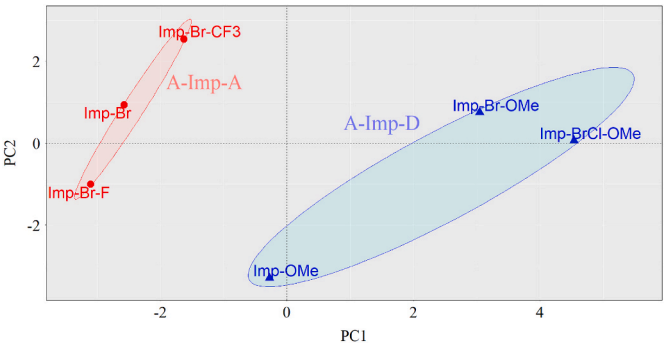


Fig. 6. Score plot for the PCA model. A-Imp-A and A-Imp-D structures were used as categorical variables and identified as red and blue dots. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

physical parameters. Notably, there is a negative correlation with the anisotropy coefficient (r) and strong positive correlations with the cubic molecular radius (a^3), fluorescence emission wavelength (λ_{flu}) and the first transition dipole moment (μ_{01}). This overlap in correlations makes it challenging to establish a direct influence of these parameters on the 2PA cross-section values.

The principal components in PCA are linear combinations of the original variables that are orthogonal to each other, effectively eliminating multicollinearity while preserving the majority of the information. Therefore, a second PCA model was developed using photophysical parameters of imidazo[4,5-*b*]pyridine derivatives extracted from linear and nonlinear optical results reported in Ref. [32]. Table 1 displays the photophysical parameters used for the PCA. The $\lambda_{S_0 \rightarrow S_1}^{1PA}$, $\lambda_{S_0 \rightarrow S_2}^{1PA}$ and λ_{flu} (nm) corresponds to the maximum absorption and fluorescence emission wavelengths, $\Delta\nu$ the Stokes shift, and ϕ_f (%) the fluorescence quantum yield. μ_{01} and μ_{02} represent the transition dipole moments between the ground and first or second excited states, respectively. The $\Delta\bar{\nu}/\Delta F$, a^3 and $\Delta\mu_{01}$, corresponds, respectively, to the ratio of the Stokes shift ($\Delta\bar{\nu}$) by Onsager factor (ΔF), the cubic molecular radius, and the difference of the permanent dipole moment, which are extracted from the solvatochromic process on the absorption and emission.

Since the variables in our dataset have different units and scales, data standardization was applied. Standardization ensures that all variables contribute equally to the analysis by transforming the dataset to have a mean of zero and a standard deviation of one. Further details about the standardization process, including the mean and standard deviation values, can be found in the Supplementary Information.

The scree plot and the covariance matrix (data not shown), corresponding to the data in Table 1, indicate that the first two principal components account for 82 % of the explained variance, meeting our initial criteria for validating the PCA model. The score plot in Fig. 6

Table 1
Photophysical parameters obtained in the linear and nonlinear characterization [32].

	Imp-Br	Imp – Br-F	Imp-Br-CF ₃	Imp-OME	Imp-Br-OME	Imp-BrCl-OME
$\lambda_{S_0 \rightarrow S_1}^{1PA}$ (nm)	330	331	331	331	338	342
$\lambda_{S_0 \rightarrow S_2}^{1PA}$ (nm)	280	281	281	279	286	287
λ_{flu} (nm)	370	385	383	380	397	409
r	0.32	0.32	0.30	0.09	0.18	0.14
ϕ_f (%)	2	18	1	66	25	34
$\Delta\nu$ (nm)	40	54	52	49	59	67
μ_{01} (D)	6.0 ± 0.6	4.5 ± 0.4	6.2 ± 0.6	7.0 ± 0.7	7.4 ± 0.7	7.6 ± 0.7
μ_{02} (D)	7.0 ± 0.7	5.9 ± 0.6	7.5 ± 0.7	6.0 ± 0.6	7.0 ± 0.7	6.8 ± 0.6
$\Delta\bar{\nu}/\Delta F$ (cm ⁻¹)	775	65	98	3870	4480	4660
a^3 (Å ³)	110.6	112.4	121.2	112.5	119.7	124.6
$\Delta\mu_{01}$ (D)	4.1	1.2	1.5	9.3	10.3	11.8

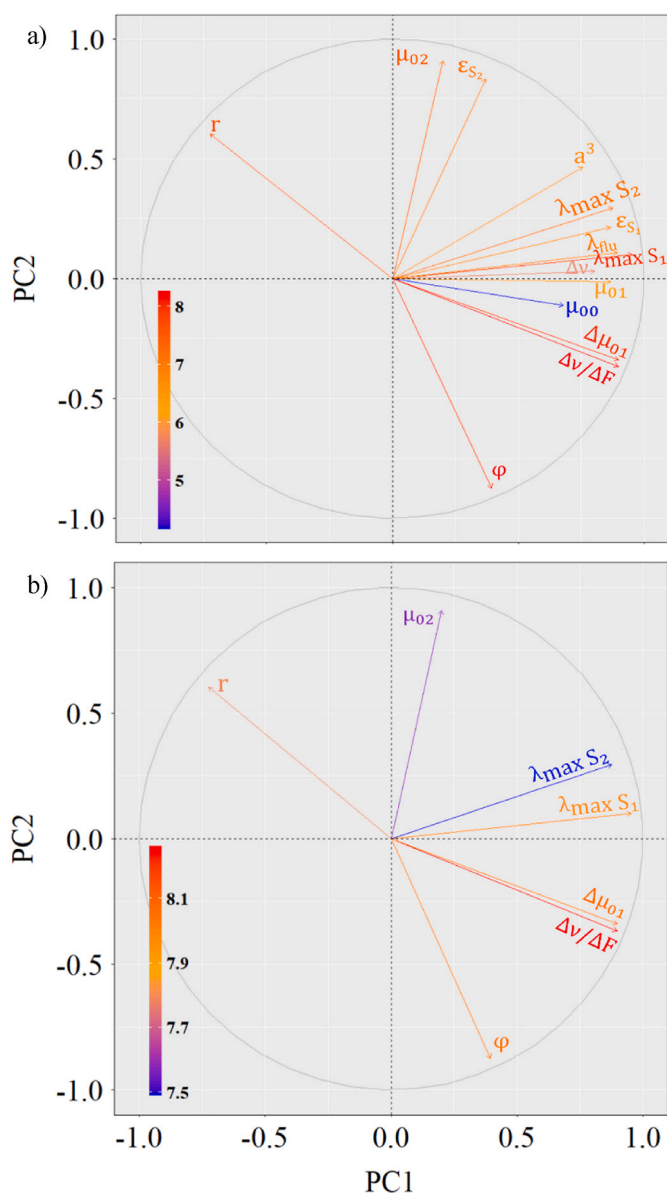


Fig. 7. a) Loading plot for the photophysical parameters characterized. b) Loading plot without the variables with less contribution than 30 % of data variability. Descriptor variables are categorized according to the variable contribution (gradient scale).

illustrates the distribution of the molecules across the component space. Specifically, A-Imp-D samples are mainly distributed along the PC1 axes, while A-Imp-A samples are distributed along the PC2 axes. It is worth noting that the six derivatives studied are separated based on their electronic structure by a 90 % confidence ellipse. This result indicates that both PCA models effectively categorize and cluster the imidazopyridine derivatives, demonstrating the accuracy of the technique for monitoring descriptor variables.

The three A-Imp-A molecules exhibit better clustering, indicating similar photophysical properties. Although A-Imp-D molecules are also clustered, the Imp-OMe sample has a significant contribution from PC2.

The loadings plot shown in Fig. 7a reveals that almost all the descriptor variables are highly correlated and oriented toward the positive PC1 axes. Parameters such as transition dipole moment to the second excited state (μ_{02}), fluorescence quantum yield (ϕ), fluorescence anisotropy (r), and molar absorption coefficient of the second absorption band (ϵ_{s2}) are distributed along the PC2 axes. The correlations

observed in the loading plot align with the experimental results, as parameters obtained using the same method exhibit the same orientation, indicating a strong correlation between them. A clear example is the relationship between the maximum absorption and emission wavelengths and their Stokes shift ($\lambda_{s1\ max}$, λ_{flu} , $\Delta\nu$) or the relation between the second transition dipole moment and the second absorption band molar absorption (μ_{02} and ϵ_{s2}). Since these variables are implicitly related, a higher value for one directly implies a higher value for the other.

However, beyond the group analysis of the variables, the loading plot also reveals a negative correlation between the fluorescence anisotropy coefficient (r) and most of the other descriptors, except for the variables corresponding to the transition intensity to the second excited state (μ_{02} and ϵ_{s2}). Similarly, there is an almost null correlation between this group and the variables derived from solvatochromic shift measurements, such as the difference in permanent dipole moments ($\Delta\mu_{01}$) and the relationship between the Stokes shift and the Onsager factor ($\Delta\bar{\nu}/\Delta F$). Finally, an almost null correlation can also be observed between the fluorescence quantum efficiency and the descriptors related to the molecule's absorptive and emissive behavior (absorption and emission wavelengths, molar absorptivities, etc).

Performing PCA with highly correlated variables is challenging, as the interpretation of the PCA is dominated by variable redundancy, making it difficult to determine which variables contribute significantly to a given component [39]. To avoid this effect, different strategies can be employed [40,41]. The variable contribution was employed as a complementary parameter to determine which variables were essential in constructing the principal components [41]. Fig. 7b displays a second loading plot, filtering out variables that contribute less than 30 % of the total variability of the data. This plot shows that the maximum absorption wavelengths for the first and second excited states ($\lambda_{max}(S_1)$, $\lambda_{max}(S_2)$), fluorescence quantum efficiency (ϕ), transition dipole moment to the second excited state (μ_{02}), Stokes shift relation with the Onsager factor ($\Delta\bar{\nu}/\Delta F$) and the difference of permanent dipole moment for the first excited state ($\Delta\mu_{01}$) are the most influential variables for the principal components construction.

The correlation Biplot, also known as Biplot graphic display, in Fig. 8, allows for comparing variable and object orientation in the score and filtered loading plot. It can be observed that Imp-Br-OMe and Imp-BrCl-OMe have a partially null correlation with the variables μ_{02} and ϕ . These two samples also exhibit a negative correlation with the anisotropy coefficient (r), but a positive correlation with the maximum absorption wavelengths and the transition and permanent dipole moments. This indicates that these samples present higher values for these parameters. The Imp-OMe sample, on the other hand, only shows a positive correlation with the ϕ . Lastly, the A-Imp-A structures display a positive correlation only with the anisotropy coefficient but a negative correlation with the other variable descriptors [42]. All these findings align with the experimental results summarized in Table 1, indicating a good correspondence between the original and the transformed data

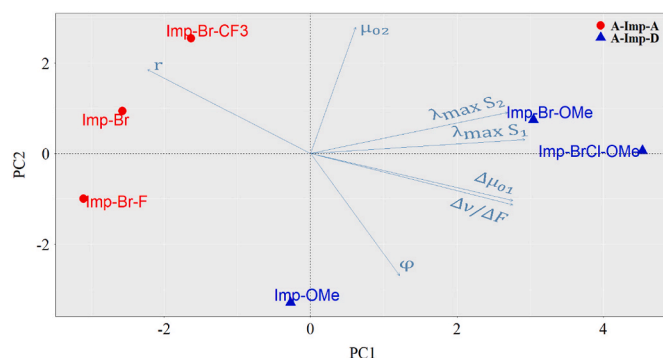


Fig. 8. Biplot graphic for samples and variables.

after the PCA [19,32].

From the results discussed before, we were able to discuss the correlation of photophysical parameters with different molecular structures. However, we did not establish a direct relationship between the photophysical parameters and 2PACS. This is because PCA utilizes orthogonal transformations to construct the principal component space, which only allows for the reduction of data dimensionality. As a result, the maximum variability of the original data is preserved in the distributions of samples and variables. However, the values of each principal component can be treated as dependent variables that can be used in a linear regression model, similar to the Principal Components Regression method [31]. This approach allows studying the correlation between the target variable (2PACS) and the principal components projections. Therefore, for a specific 2PACS value, the PC value, whether high or low, can be interpreted as the influence of the photophysical parameters of a given molecule on this nonlinear property. Principal components and 2PA cross-section for all molecules are summarized in Table 2.

The correlation two-dimensional plots of the PC1 and PC2 scores, for each derivate, concerning 2PACS is shown in Fig. 9. The color scale corresponds to the range of the 2PACS values obtained (79–130 GM) [32].

Linear regression analysis reveals that PC1 plays a dominant role over PC2, offering a more accurate description of the molecular behavior. The sample distribution in the PC1 correlation plot shows a stronger agreement with the experimental 2PACS values, as molecules from the A-Imp-D group exhibited higher 2PACS values (as shown in Fig. 9a). Conversely, the A-Imp-A group samples displayed lower 2PACS values, indicating a negative correlation and a limited influence of PC1 for this group.

From the results in Figs. 7 and 8, and the regression models (Fig. 9a and b), it can be inferred that PC1 has a greater influence on samples with higher 2PA cross-section values. This suggests that variables highly correlated with PC1, such as transition dipole moment (μ_{01}), permanent dipole moment change ($\Delta\mu_{01}$), and the Stokes shift normalized by the Onsager polarizability ($\Delta\nu/\Delta F$), significantly impact the 2PACS. Furthermore, a lower anisotropy coefficient (r) also contributes positively to higher 2PACS values. However, it is important to note that μ_{02} and fluorescence quantum efficiency (ϕ) exhibit negligible correlation with PC1, as evidenced by Table 1, where A-Imp-D molecules maintain favorable values for these parameters despite their limited contribution to PC1. Interestingly, the PC2 distribution adds a complementary layer of information. For example, the Imp-CF₃ and Imp-OMe samples, which exhibit the highest and lowest PC2 projections, respectively, show a positive correlation with μ_{02} and ϕ . This observation indicates that these variables play a more significant role in influencing the 2PACS values in these cases. Expanding the analysis to variables not highlighted in the filtered loadings plot (Fig. 7), it can be inferred that the intensity of the absorption band for the second excited state (ϵ_{S2}) and the cubic radius (α^3) also influence the 2PACS value. This is supported by their high contributions to PC2 and PC1, respectively.

The results discussed above are consistent with previous reports in the literature, where an increase in molecular nonlinearity is directly associated with an intense charge transfer process, reflected in higher transition dipole moments [43,44]. Similarly, a redshift in absorption peaks, indicative of lower energy gaps, is often correlated with enhanced

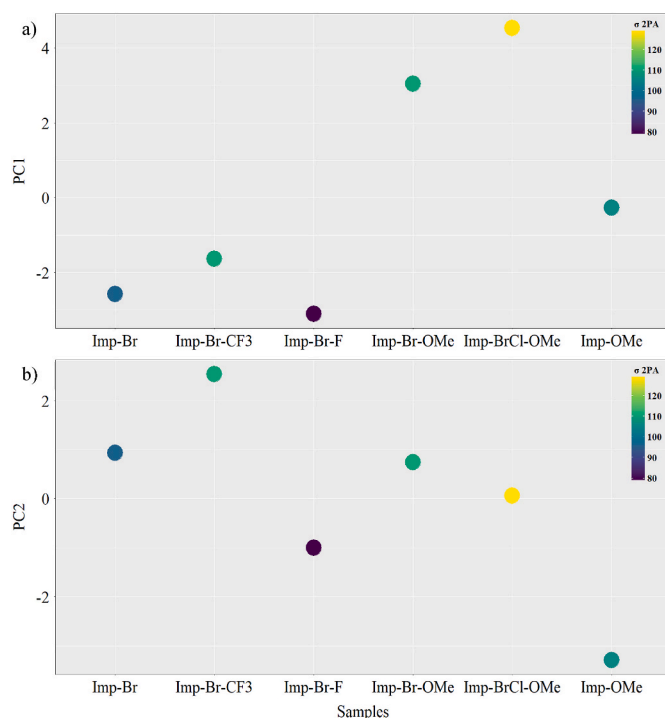


Fig. 9. Linear regression of the two-photon absorption cross-section value with (a) PCA1 and (b) PCA2 for all the Imidazo[4,5-*b*]pyridine derivatives.

NLO responses due to more easily accessible excited states. Furthermore, high fluorescence quantum efficiency suggests that the molecule efficiently returns to its ground state through radiative pathways, minimizing non-radiative decay. This behavior is typically associated with stable excited states and an enhanced two-photon absorption process. Finally, these parameters collectively highlight the significant role of molecular structure, particularly electron donor-acceptor interactions. Planar systems with extended π -conjugation and strong donor-acceptor interactions tend to exhibit greater nonlinearity, further enhancing their optical properties [43,44].

4. Conclusion

Principal Component Analysis was used as an exploratory technique to model the linear and nonlinear properties of six derivatives of Imidazo [4,5-*b*] Pyridine. Similarly, Linear Regression Modeling was employed to evaluate the correlation between the two-photon absorption cross-section and the photophysical parameters obtained during the characterization of both linear and nonlinear optical properties of the Imidazo [4,5-*b*]pyridine derivatives. The PCA model effectively demonstrated significant clustering of the Imidazo[4,5-*b*]pyridine samples, based on their electronic structure. Furthermore, the distribution of variables displayed a high correlation among nearly all descriptor variables, with good data variability as denoted by the variable contribution.

The regression analysis shows that PC1 provides a better distribution of molecules as a function of the experimental cross-section values. Specifically, PC1 has a greater influence on the molecules of the A-Imp-D group. While, for the A-Imp-A samples, PC2 dominates over PC1 exhibiting a greater influence. The first principal component shows a strong positive correlation between the magnitude of the two-photon absorption cross-section, the first excited state transition dipole moments, and the absorption and fluorescence wavelengths. Similarly, the second principal component shows the same pattern for the fluorescence anisotropy, the second excited state transition dipole moment, and the fluorescence quantum yield. These positive correlations indicate that these variables mainly describe the 2PACS behavior, indicating that,

Table 2
Parameters for the Principal component regression and 2 PA cross-section.

	Imp-Br	Imp-Br-F	Imp-Br-CF ₃	Imp-OMe	Imp-Br-OMe	Imp-BrCl-OMe
σ_{2PA}	100 ± 10	79 ± 8	110 ± 10	110 ± 10	110 ± 10	130 ± 10
(GM)	10			10	10	
PC1	-2.572	-3.102	-1.631	-0.274	3.045	4.534
PC2	0.936	-0.999	2.543	-3.288	0.747	0.061

samples with higher values of positively correlated variables will tend to have higher 2PACS values.

These findings align closely with previous studies in the literature, which indicate that the nonlinear optical properties of molecules are significantly influenced by their electronic structure. For example, an intense charge transfer process, indicated by higher transition dipole moments, is a crucial factor that contributes to increased nonlinearity. Additionally, a redshift in absorption peaks is often associated with enhanced two-photon absorption processes. High fluorescence quantum efficiency further promotes effective radiative transitions. Ultimately, all these factors are intrinsically related to donor-acceptor interactions in molecules with planar structures and extended π -conjugation, which play a pivotal role in amplifying their nonlinear optical responses.

CRedit authorship contribution statement

O.D. Marbello: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **L. De Boni:** Writing – review & editing, Validation, Supervision. **C.R. Mendonça:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.optmat.2024.116620>.

Data availability

No data was used for the research described in the article.

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